

# Peter Keel

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## EXPERIENCE

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**DRW** Chicago, IL  
*Market Data Systems Engineer* Apr 2024 – Present

- Built a continuous futures roll system used daily by a trading desk to determine when and how to roll positions across 190+ products and 36 exchanges
- Curated and validated historical market data for 300+ researchers, traders, and quant modelers, including symbology mapping, daily data quality investigations, and conversion of archives to Parquet for faster access
- Led the migration and versioning of research databases across the firm, coordinating across trading, risk, and research teams to establish reproducible data architecture
- Built an options analytics layer providing implied volatility surfaces, Greeks, and strike price scaling across CME, ICE, and Eurex, and developed corporate actions and dividend forecast pipelines used in pricing models
- Extended a vendor time-series library with point-in-time query functionality and built an AI-powered data query assistant adopted firm-wide
- Serve as primary point of contact for the firm's market data platform, leading demos and onboarding sessions for trading, risk, and research teams

**Cboe Global Markets** Chicago, IL  
*Product Manager* Nov 2021 – Apr 2024

- Led development of an automated NewsFeed from concept to product, overseeing sprints with developers, specification writing, and API documentation in Python and Jinja
- Developed Python scripts automating quarterly index rebalancing validations, saving hours of recurring work for the Cboe index team
- Supervised an intern through a specialized project, conducting regular meetings and organizing a final presentation with group leaders

**Nasdaq** Rockville, MD  
*Senior Index Analyst* Apr 2021 – Nov 2021  
*Index Analyst* Apr 2020 – Apr 2021

- Converted 20+ legacy Access/Excel index models into Python, resulting in 3x faster execution and 30-80% reduction in reconstitution time
- Built daily automated quality checks detecting discrepancies in index levels, constituent changes, and corporate actions
- Performed analysis and mapping for Bloomberg to Intercontinental Exchange fixed income data migration

**Wells Fargo** Minneapolis, MN  
*Data Science Intern* Dec 2019 – Jan 2020

- Built a Python web app for traders to receive bond orders from portfolio managers, prioritizing bonds with favorable in-house credit ratings

**Piper Jaffray Companies** Minneapolis, MN  
*Analytics Fixed Income Intern* Jun 2019 – Dec 2019

- Developed statistical models to assign credit scores to municipalities, predict corporate bond rating changes, and predict defaults using data cleaning, parameter tuning, and train/test validation

## PROJECTS

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**U.S. Housing Market Explorer** 2026

- Built an interactive geospatial dashboard visualizing housing data across 894 U.S. metros from 8 data sources (Zillow, Census ACS, FRED, BEA, FEMA, EPA). Automated weekly data refresh via GitHub Actions. [Live demo](#)

**NBA Hall of Fame Prediction Model** 2020 / 2026

- Modeled NBA Hall of Fame induction probability for 380+ players using random forest with 10-fold cross-validation (94.7% accuracy). Built in R/Shiny (2020), rebuilt in Python/Streamlit (2026). [Live demo](#)

## SKILLS

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**Coding:** Python (Pandas, Scikit-learn, SciPy, Plotly), SQL, R, Bash, JavaScript, HTML/CSS

**Data & Tools:** OneTick, Bloomberg, Parquet, Trino, Databricks, AWS, Terraform, Docker, Airflow, Git, Linux

## EDUCATION

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**Carleton College** Northfield, MN  
*Bachelor of Arts in Mathematics, GPA: 3.64* Class of 2020

- **Relevant Coursework:** Real Analysis, Multivariable Calculus, Linear Algebra, Statistical Inference, Applied Regression, Bayesian Statistics, Statistical Consulting, Combinatorics, Data Science, Data Structures, Programming Languages
- **Extracurricular:** Varsity Soccer & Track (Academic All-Conference), Carleton Investment Office Intern, Math Tutor

**Dartmouth Tuck School of Business** Hanover, NH  
*Bridge Program* Dec 2018